

BDM CAPSTONE VIVA • IIT MADRAS ONLINE BS

Optimizing Inventory Management & Sales in a Retail Pharmacy

Data-Driven Techniques & Strategies for Ganesh Medicals

Presented by

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Ganesh Medicals

Bajla Chowk, Deoghar, Jharkhand



Organization Background: Ganesh Medicals



Overview

- B2C retail pharmacy at Bajla Chowk, Deoghar, Jharkhand
- Serves walk-in & prescription customers; **766 SKUs**; 65,343 transactions
- Average monthly revenue $\approx$ **₹16.53 lakh** during study window



Data & Operations

- Digital POS & inventory system ensuring reliable primary data
- Study: Sales Apr 1–Oct 5 (188 days); Proc. Apr 1–Nov 12 (226 days)
- Procurement volume: 118,312 units; free units $\approx$ **20.1%**



Key Challenges

- Maintaining optimal stock while minimizing expiry-linked losses
- Managing seasonality-driven demand swings and surge cycles
- Mitigating supplier variability and complex procurement constraints

Problem Statements

Core operational challenges identified at Ganesh Medicals



01

Inventory Inefficiency & Expiry-Linked Losses

Lack of data-driven planning leads to overstocking, stockouts, and returns. Slow movers pose a severe expiry risk.

BASELINE EVIDENCE

Returns: ₹1.78 lakh (2.75% of units)

Stockouts: 0.43% of true demand

❗ 2 products at 100% return rate



02

Seasonality-Driven Demand Misalignment

Procurement is not fully aligned with seasonal swings, causing missed sales during surges and excess stock post-peak.

BASELINE EVIDENCE

Sales Surge: +54.6% jump (Jul→Sep)

Alignment: MoM correlation $r = 0.86$

⚠ Timing gaps during transition phases



03

Supplier Variability & Procurement Constraints

Irregular deliveries and complex free-qty schemes distort the true cost of goods, creating inefficiencies.

BASELINE EVIDENCE

Rate Spread: ₹53.25 to ₹100.78/unit

Cost Difference: 89% spread

✅ Identified clear cost-leaders

Data & Methodology



1. Primary Data Collection

- **Sales Data:** 65,343 transactions spanning 188 days (Apr 1–Oct 5, 2025)
- **Procurement Data:** 118,312 units over 226 days (Apr 1–Nov 12, 2025)
- **Promotional Schemes:** Free units account for **≈20.11%** of total volume
- **Scope:** 766 unique SKUs directly exported from POS and inventory systems



2. Data Cleaning & Preparation

- Removed 187 daily aggregate rows to ensure transaction-level precision
- Standardized naming conventions (e.g., TAB/TABLET, INJ/INJECTION) to match sales and purchase sets
- Segregated negative quantities to isolate true **returns (S/Re)** from unmet demand and **stockouts (shortfall)**



3. Analytical Tools

- **Microsoft Excel:** Used for primary data cleaning, normalization, metadata documentation, and ABC classification
- **Python (Pandas, Matplotlib):** Used for sales velocity computation, MoM tracking, effective rate modeling, and visualizations



4. Methods Mapped to Problem Statements

PS1: Inventory Inefficiency & Expiry Losses

Applied **ABC + Sales Velocity Matrix** to segment products.

$$\text{Stockout Impact} = |\text{Stockout}| / (\text{Sales} + |\text{Stockout}|)$$

PS2: Seasonality Misalignment

Conducted **Month-over-Month (MoM) Growth Comparison**.

$$\text{True Demand} = \text{Sales Qty} + |\text{Documented Stockouts}|$$

PS3: Supplier Variability

Multi-criteria **Supplier Performance Scorecard** integration.

$$\text{Effective Rate} = (\text{Rate} \times (1 - \text{Discount\%}) \times \text{Qty}) / (\text{Qty} + \text{FreeQty})$$

Key Finding 1 - Inventory Inefficiency & Expiry-Linked Losses

REVENUE CONCENTRATION

22% SKUs = 80% Rev

169 items generate ₹79.4L of ₹99.2L

TOTAL RETURN IMPACT

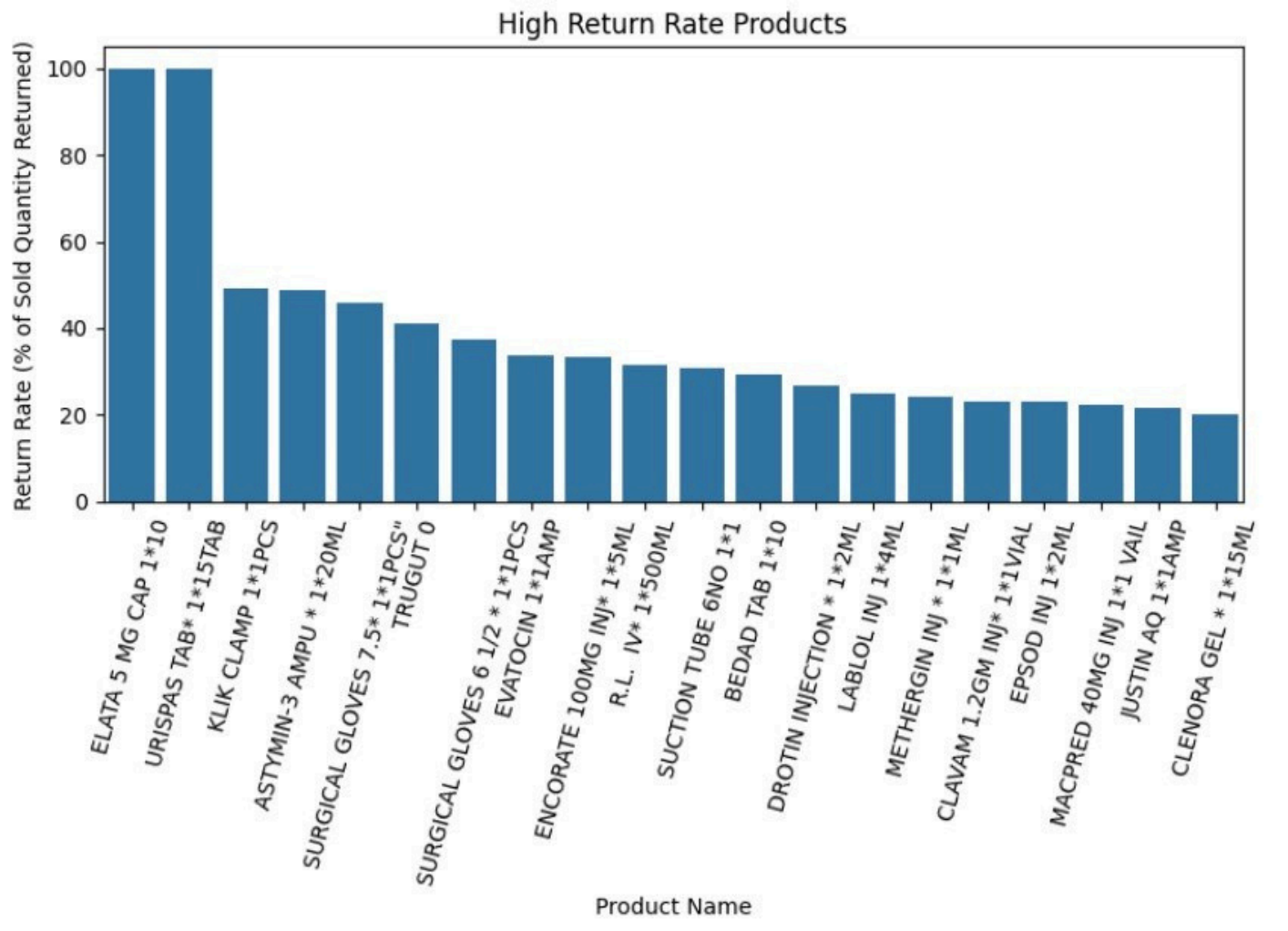
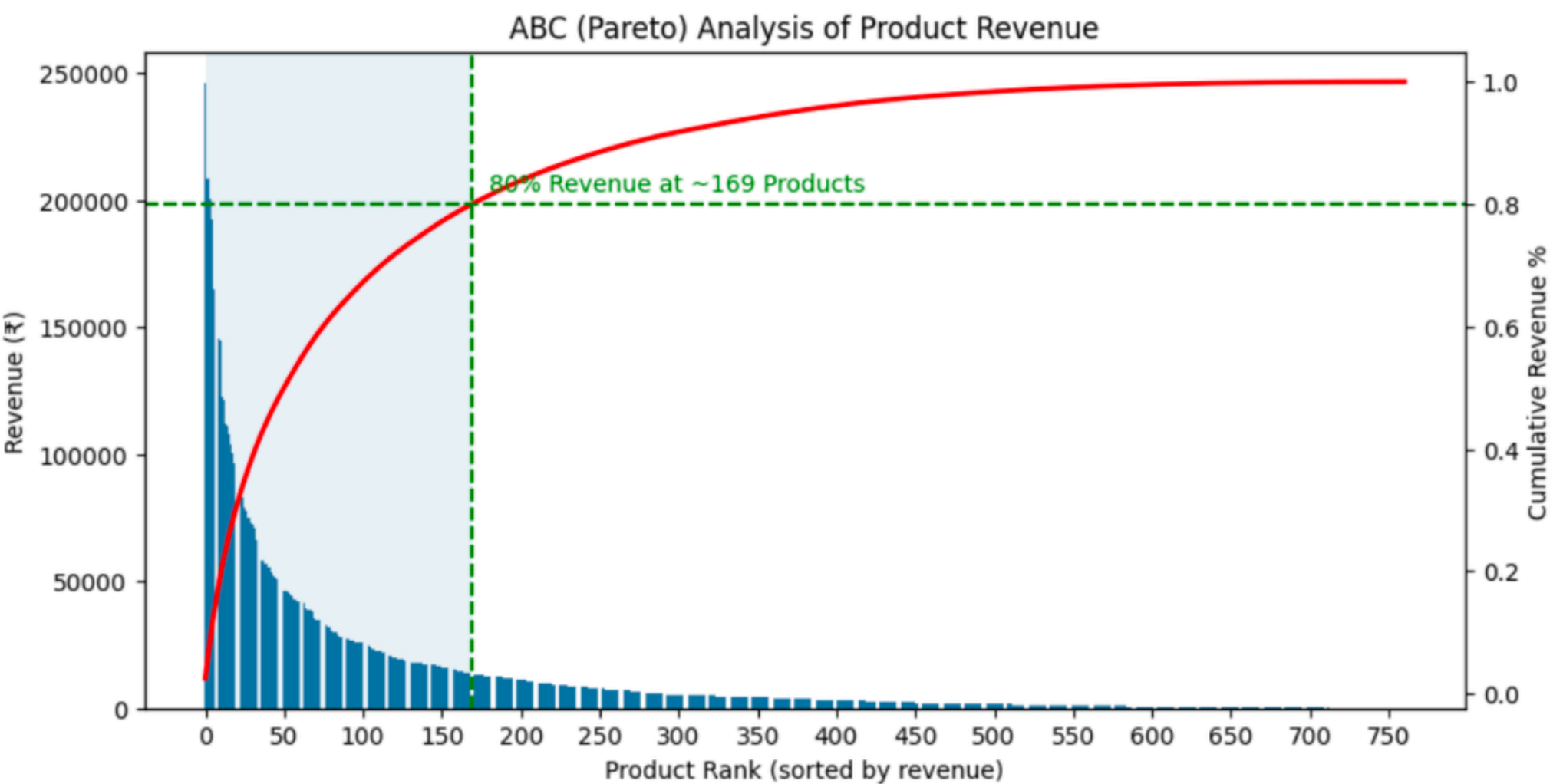
₹1.78L (2.75%)

2,914 units returned

TRUE DEMAND LOSS

0.43% Stockout Rate

Missed sales due to unavailability



💡 **Insight:** Long-tail slow movers drive expiry risk. 2 SKUs hit 100% returns; surgical items average 30–49% returns.

Key Finding 2 - Seasonality-Driven Demand Misalignment

DEMAND SURGE

54.6% Increase

Sales jump from July dip to Sept peak

ALIGNMENT STRENGTH

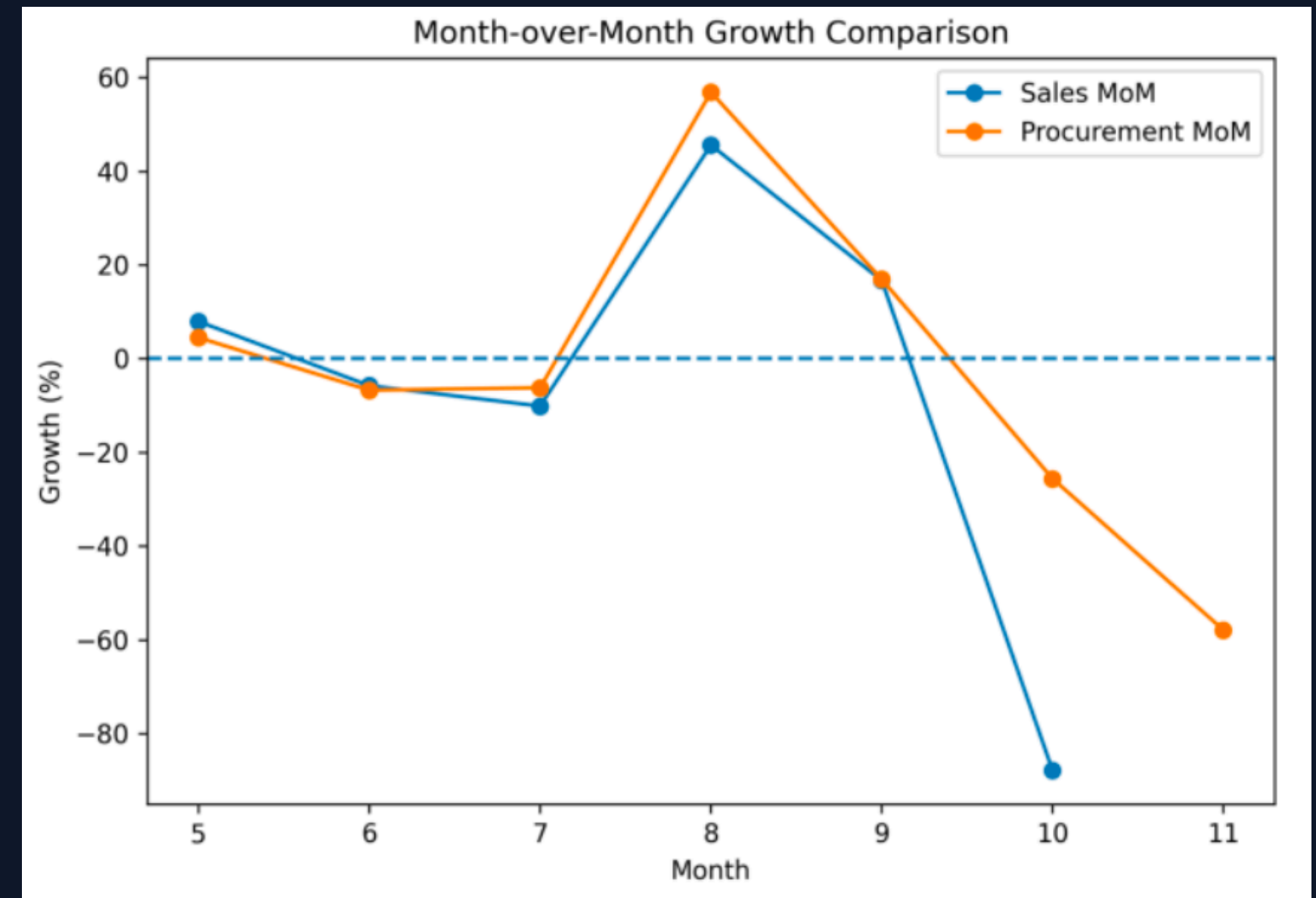
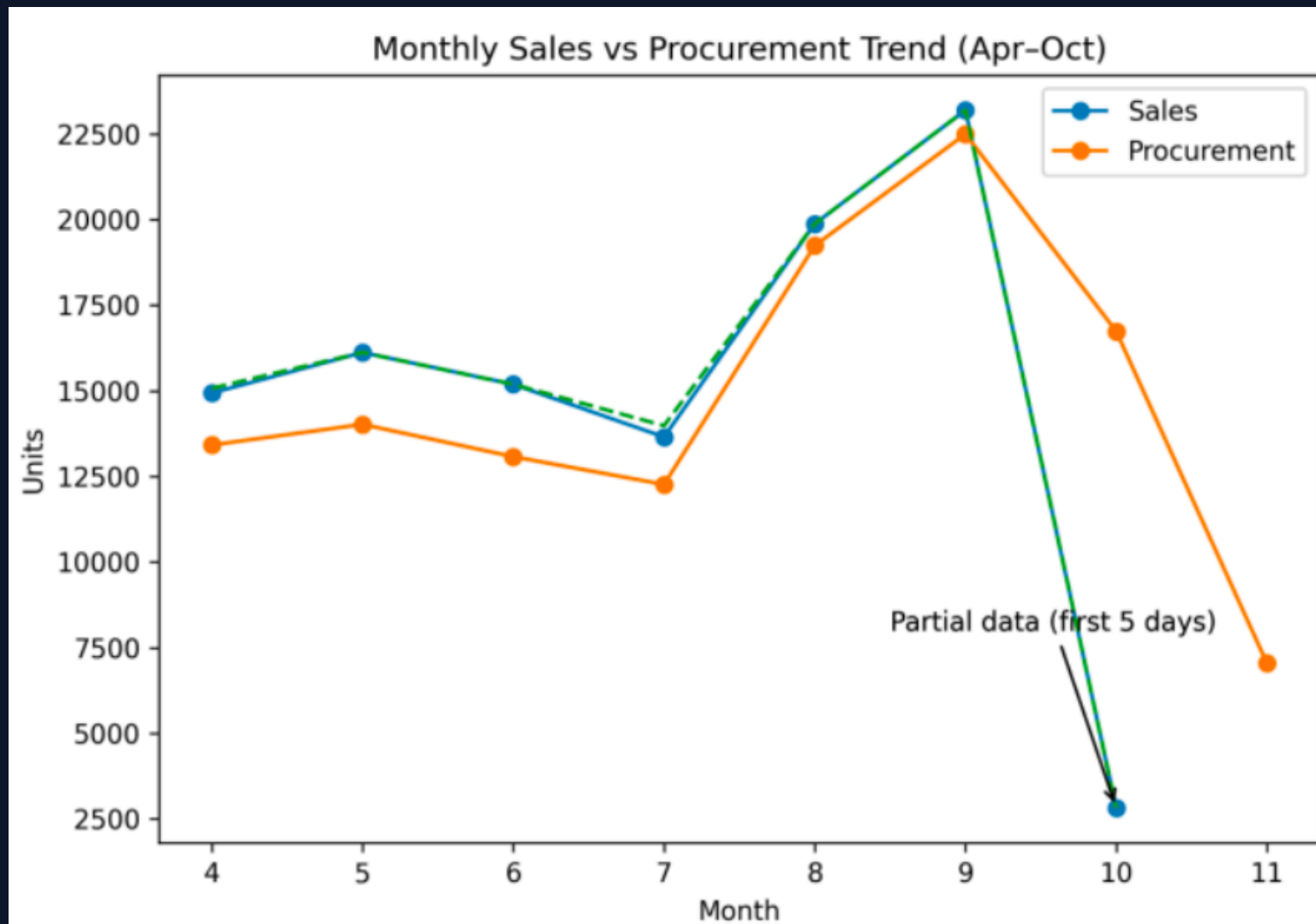
$r = 0.86$

Sales vs Purchase MoM Correlation

AUGUST TRANSITION GAP

3.5% Lag

Procurement (+42.1%) vs Sales (+45.6%)



💡 **Insight:** Procurement decisions follow demand directionally but lag during rapid transitions (August surge), risking stockouts.

Key Finding 3 - Supplier Performance Analysis

UNIFORM HEADLINE DISCOUNTS

~4.75% – 5.0%

Masks actual cost differences between vendors

COST VARIANCE SPREAD

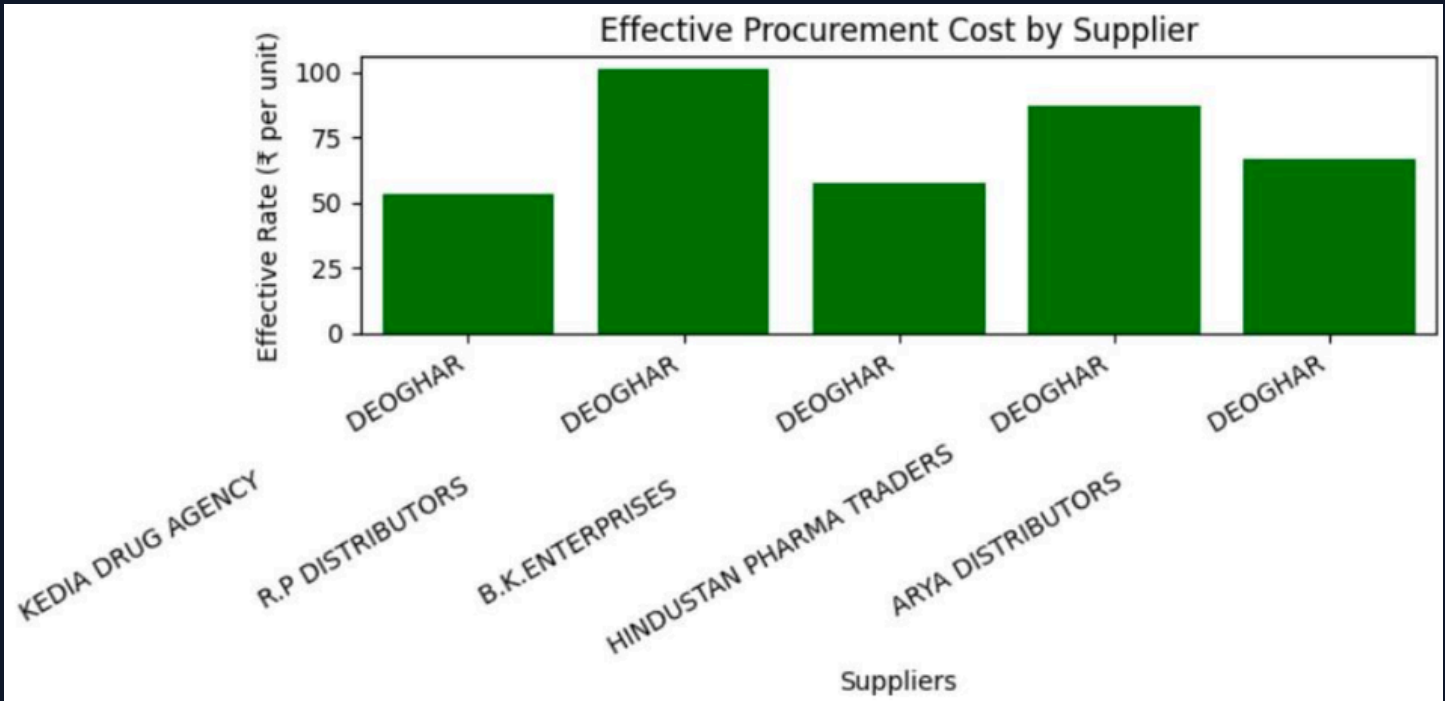
89% Difference

From ₹53.25 (Kedia) to ₹100.78 (R.P. Dist)

POTENTIAL ANNUAL SAVINGS

₹16.05 Lakh

By shifting 118,312 units to low effective-rate suppliers



💡 **Insight:** B.K. Enterprises offers highest overall value. Selecting vendors by Effective Rate (incorporating free quantities) uncovers massive cost advantages over flat discount rates.

Recommendations

Strategic actions mapped directly to the three core operational challenges



PS1

Inventory Inefficiency & Expiry Losses

STRATEGIC ACTIONS

- **ABC + Velocity Policy:** Implement strict availability control for 169 A-category high-value items.
- **Reorder Points:** Establish dynamic levels based on daily velocity $\times$ lead time; enforce FEFO.
- **Stock Rationalization:** Prune slow/low-value SKUs and minimize high-return items.
- **On-Demand Sourcing:** Shift surgical/specialty items to order-based procurement.



PS2

Seasonality-Driven Demand Misalignment

STRATEGIC ACTIONS

- **Seasonal Forecasting:** Anticipate Aug-Sep peaks; apply 1.4x multipliers for monsoon-related meds.
- **Advance Procurement:** Build stock 4–6 weeks before anticipated demand surges.
- **True Demand Planning:** Use Sales + |Stockouts| instead of just historical sales data.
- **Dynamic Cadence:** Abandon rigid monthly ordering in favor of responsive schedules.



PS3

Supplier Variability & Procurement Constraints

STRATEGIC ACTIONS

- **Effective Rate Selection:** Choose vendors based on actual unit cost, not nominal discount %.
- **Volume Shifting:** Consolidate purchases with Kedia, B.K., and Arya to maximize price competitiveness.
- **Renegotiation:** Address high-cost suppliers (e.g., R.P. Distributors) armed with scorecard data.
- **Supplier Diversification:** Expand network to reduce dependency risks.

Expected Impact - Financial and Operational

TOTAL ANNUAL SAVINGS

₹8.1L – ₹11.6L

Projected bottom-line impact

MARGIN IMPROVEMENT

+11% to +15%

Increased profitability & efficiency

RETURN LEAKAGE CONTROL

Target: ≤ ₹0.9L

Eliminate all 100%-return SKUs

Projected annual savings breakdown



■ Procurement optimization
₹5.8 – 8.7L

■ Seasonal demand capture
₹1.5 – 2.0L

■ Reduced returns / expiry
₹0.8 – 0.9L

Key Performance Indicator (KPI) Targets

📦 Stockout Rate **0.43%** → **≤ 0.20%**

↺ Return Leakage **₹1.78L** → **≤ ₹0.90L**

✓ A-Item Availability **Variable** → **≥ 98.0%**

🚀 **Transformation:** Transitioning Ganesh Medicals from reactive, heuristic-based operations to predictive, analytically-supported inventory management.

Thank You!



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🏛 IITM Online BS Degree Program



SPECIAL THANKS TO

Ganesh Medicals

Mr. Kedia (Owner)

📍 Bajla Chowk, Deoghar, Jharkhand

For providing necessary resources and authorized access to business data.